

Xinsong Feng

Ph.D. student in Data Science working on reinforcement learning, generative models, and language models, with a current focus on diffusion language models and efficient inference.

(+1) 213-791-1673
xfeng06@wm.edu
xfeng01.github.io

Education

William & Mary

Ph.D. in Data Science, GPA: 4.00/4.00, Advisor: Haipeng Chen

Williamsburg, VA, USA

Aug. 2025–Present

University of California, Los Angeles

M.S. in Electrical and Computer Engineering, GPA: 4.00/4.00

Los Angeles, CA, USA

Sep. 2023–Jun. 2025

Chongqing University

B.Eng. in Communication Engineering, Major GPA: 3.94/4.00

Chongqing, China

Sep. 2019–Jun. 2023

Research Experience

Data-Driven Decision Intelligence (D3I) Lab, William & Mary

Ph.D. Student Researcher

Aug. 2025–Present

Advisor: Haipeng Chen, Assistant Professor

DELA: Controllable Diffusion Language Models via RL-based Editing

Oct. 2025–Present

- Developed DELA, an RL-based framework that trains an editor policy to control diffusion language model generation through iterative editing of partial sequences.
- Designed token-level edit actions and policy objectives for revising intermediate generations, enabling more flexible control over sequence structure and decoding behavior.
- Built a controllable generation pipeline for diffusion language models with a focus on generation quality, structured editing, and inference efficiency.

Constrained Optimization via RL and ODE-based Models

Mar. 2025–Sep. 2025

- Proposed Constrained Markov Flow Optimizer (CMFO), a framework unifying reinforcement learning and ODE-based generative models for constrained optimization.
- Formulated constrained optimization as sequential decision-making over iterative refinement dynamics, establishing a connection between policy learning and ODE-based optimization.
- Introduced RelaxCompletion for differentiable feasibility handling and achieved state-of-the-art performance with substantially faster inference than classical solvers.

University of California, Los Angeles

M.S. Student Researcher

Sep. 2023–Jun. 2025

Research collaboration with Prof. Haipeng Chen

Offline Reinforcement Learning with Generative Trajectory Policies

Aug. 2024–May 2025

- Proposed Generative Trajectory Policies (GTP), a unified offline RL framework connecting diffusion, flow matching, and consistency models through a continuous-time trajectory formulation.
- Derived a closed-form score approximation from offline data, eliminating iterative ODE solvers and enabling stable and efficient policy learning.
- Introduced advantage-weighted value guidance and achieved state-of-the-art results on D4RL, including perfect scores on multiple AntMaze tasks.

Sequential Stochastic Combinatorial Optimization with Hierarchical RL

Oct. 2023–May 2024

- Formulated sequential stochastic combinatorial optimization as a new decision-making setting with uncertain dynamics and combinatorial action spaces.
- Developed a hierarchical RL framework with tailored state, reward, and null-action design for efficient learning over large combinatorial action spaces.
- Proposed a wake-sleep option training strategy and outperformed strong baselines by 20% on adaptive influence maximization and route planning.

Publications

- [1] H. Meng, **X. Feng**, et al. "Solving Constrained Optimization Problems as ODE-based Models Using Reinforcement Learning," submitted to *TMLR*.
- [2] **X. Feng**, L. Tang, C. Wang, and H. Chen. "Offline Reinforcement Learning with Generative Trajectory Policies," *ICML*, 2026. *arXiv preprint* arXiv:2510.11499, 2025.
- [3] **X. Feng**, Z. Yu, Y. Xiong, and H. Chen. "Sequential Stochastic Combinatorial Optimization Using Hierarchical Reinforcement Learning," *ICLR*, 2025. *arXiv preprint* arXiv:2502.05537, 2025.
- [4] B. Zhou, X. Guo, **X. Feng**, et al. "Complex-Frequency Selective Resonances in Disordered Time-Varying Cavities," *Laser & Photonics Reviews*, e01228, 2025.
- [5] **X. Feng**, I. Roberts. "Adaptive Cell Range Expansion in Multi-Band UAV Communication Networks," *arXiv preprint* arXiv:2411.18123, 2024.
- [6] B. Zhou, **X. Feng**, et al. "Transition from normal to non-normal distributions of an electromagnetic field in a disordered time-varying cavity," *Physical Review A*, 110(6), 063524, 2024.
- [7] B. Zhou, **X. Feng**, et al. "Susceptibility Invariance and Duality-Matching Condition for Perpendicular-Motion Metasurface," *Advanced Physics Research*, 3(10), 2400073, 2024.
- [8] Y. Wei, L. Liang, B. Zhou, and **X. Feng**. "A Modified Blockchain DPoS Consensus Algorithm Based on Anomaly Detection and Reward-Punishment," *IEEE International Conference on Communication Software and Networks*, 283–288, 2021.

Skills

Programming: Python, C++, MATLAB

ML / LLM Frameworks: PyTorch, Hugging Face Transformers, NumPy, Pandas, Scikit-learn

Research Areas: Reinforcement Learning, Offline RL, Diffusion Models, Generative Modeling, Controllable Generation

Systems & Tools: Linux, Git, Docker, Kubernetes

Teaching Experience

Graduate Teaching Assistant, *University of California, Los Angeles*

Spring 2024

Course: *Introduction to Communication Systems*. Graded assignments and exams and provided feedback to support student learning.

Undergraduate Teaching Assistant, *Chongqing University*

Fall 2021–Fall 2022

Courses: *Digital Signal Processing* (Fall 2021, Fall 2022); *Introduction to Programming with C++* (Fall 2022). Supported lectures and office hours, prepared course materials, and graded homework and exams.

Selected Honors and Awards

Distinguished Student, Chongqing University (top 10/266 in department)

May 2023

Top 10 Outstanding Teaching Assistant, Chongqing University

Jan. 2023

National Scholarship for Academic Excellence (top 4/266 in department)

Dec. 2022

Outstanding Graduate, Chongqing University (top 25/266 in department)

Nov. 2022

First Prize, Chinese Mathematics Competitions (twice; top 0.2% nationally)

Dec. 2020, Dec. 2021

Professional Service

Reviewer, AISTATS 2026

Oct. 2025

Reviewer, IEEE GLOBECOM 2024

Jun. 2024